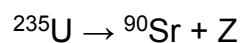


Nuclear medicine uses radiopharmaceuticals for disease treatment and as tracers in medical imaging studies. Yttrium-90 (^{90}Y) is a radiopharmaceutical agent used to treat overgrown joint lining known as pigmented villonodular synovitis, as well as some forms of liver cancer. It has a half-life of about 64 hours.

Although the relatively short half-life of ^{90}Y is optimal for use in medical applications, transportation and storage of the radiopharmaceutical are not feasible. As a result, strontium-90 (^{90}Sr), with a half-life of 28.8 years, is the most common source of ^{90}Y . ^{90}Sr is created by a nuclear fission process that begins with uranium-235 (^{235}U) in a nuclear reactor as shown in Reaction 1.



Reaction 1

Fission of ^{235}U produces ^{90}Sr as well as additional fission products (Z) including, but not limited to, technetium-99 (^{99}Tc), iodine-129 (^{129}I), and zirconium-93 (^{93}Zr). Figure 1 shows the distribution of fission products.

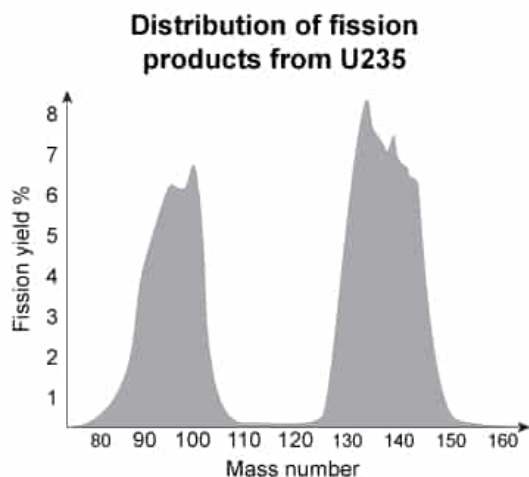
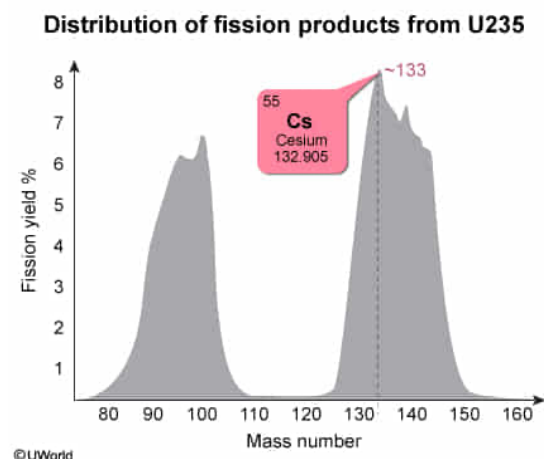


Figure 1 Distribution of ^{235}U fission products by mass number

To create the final ^{90}Y required for nuclear medicine studies, ^{90}Sr decay is carried out in a controlled ^{90}Y generator. ^{90}Y is then separated from residual ^{90}Sr for use in clinical applications.

Fission of ^{235}U produces multiple elements. Based on Figure 1, which of the following is the atomic number of the most common fission product?

- ☐ A. 38 [5%]
- ✓ ☒ B. 55 [45%]
- ☐ C. 92 [5%]
- ☐ D. 133 [43%]

Explanation:

Nuclear fission is the splitting of a large atomic nucleus into **multiple smaller nuclei**. Figure 1 displays the **mass numbers** of the various nuclear fission products for ^{235}U on the x-axis and the percentage of nuclei with each mass number on the y-axis. The tallest peak indicates the greatest fission yield (ie, the percentage of the isotope released in the greatest quantity).

On this graph, the greatest fission yield corresponds to an isotope with a mass number of approximately 133. The **periodic table** displays the average atomic weight (ie, the combined masses of the protons and neutrons) of each element under the element's symbol, calculated from the natural abundance of each possible isotope for that element. It shows that the elements iodine (I) through barium (Ba) have average atomic weights near 133. Of the answer choices given, only cesium (Cs) with an **atomic number** of 55 has a mass that falls within this range.

(Choice A) Strontium has an atomic number of 38 and an average mass number of about 88, likely representing one of the lower peaks of the graph.

(Choice C) The element with an atomic number of 92 is uranium. Because the question is about the fission (splitting) of uranium, all fission products must have an atomic number below 92.

(Choice D) The mass number, not the atomic number, of the most abundant fission product is 133. The mass number represents the total number of neutrons and protons; the atomic number is the number of protons only.

Educational objective:

Nuclear fission is the splitting of a large atomic nucleus into multiple smaller nuclei. The mass numbers (the sum of the protons and neutrons) of resulting nuclei can be measured and correlated with their most likely atomic numbers (number of protons).

Time spent: 0 sec

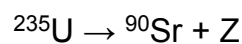
Subject:
General Chemistry

QID: 400198

System:
Atomic Theory and Chemical Composition

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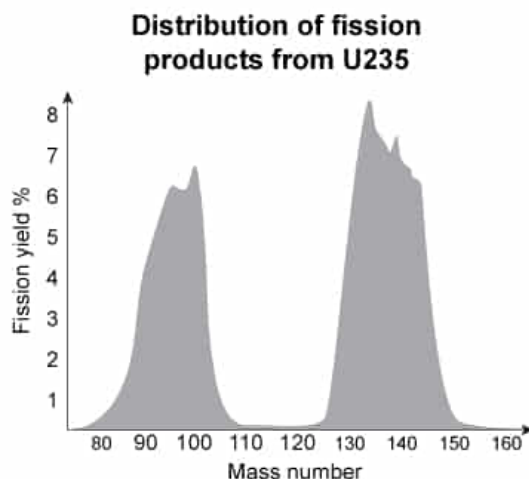


Figure 1 Distribution of ^{235}U fission products by mass number

To create the final ^{90}Y required for nuclear medicine studies, ^{90}Sr decay is carried out in a controlled ^{90}Y generator. ^{90}Y is then separated from residual ^{90}Sr for use in clinical applications.

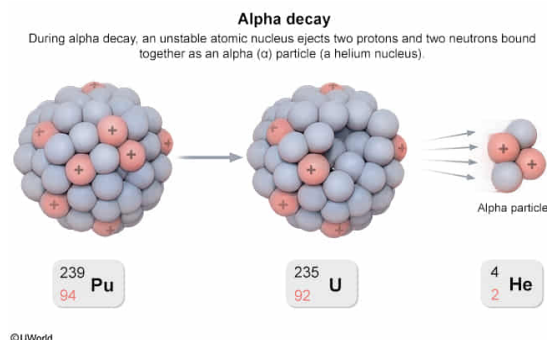
The uranium isotope required for Reaction 1 can be formed by the emission of an alpha particle from which of the following nuclei?

- ☐ A. ^{239}U [11%]
- ✓ ☒ B. ^{239}Pu [76%]
- ☐ C. ^{235}U [3%]
- ☐ D. ^{235}Pa [8%]

Correct

75% answered correctly

Explanation:



Alpha decay is the emission of an alpha particle (ie, a **helium-4 nucleus**) containing **two neutrons, two protons**, and no electrons from the parent atom. The resulting (daughter) nucleus has a **mass number** that is **four less** than the mass number of the original (parent) nucleus because the total number of protons and neutrons lost is four. The daughter has an **atomic number** that is **two less** than that of its parent because two protons are lost.

The uranium isotope required for Reaction 1 has a mass number of 235 and an atomic number of 92. Therefore, it can be generated by the alpha decay of a nucleus with a mass number of 239 (ie, $235 + 4$) and an atomic number of 94 (ie, $92 + 2$). Plutonium-239 ($^{239}_{94}\text{Pu}$) becomes uranium-235 ($^{235}_{92}\text{U}$) after alpha decay.

(Choice A) Alpha decay of $^{239}_{92}\text{U}$ leads to the formation of thorium-235 ($^{235}_{90}\text{Th}$), which has the desired mass number but an atomic number that is two less than uranium.

(Choice C) Alpha decay of $^{235}_{92}\text{U}$ leads to thorium-231 ($^{231}_{90}\text{Th}$).

(Choice D) Emission of an alpha particle from protactinium-235 ($^{235}_{91}\text{Pa}$) results in actinium-231 ($^{231}_{89}\text{Ac}$).

Educational objective:

Alpha decay is the emission of an alpha particle (helium-4 nucleus) containing two neutrons, two protons, and no electrons from the parent atom. Alpha decay results in a daughter nucleus with a mass number that is four less than its parent and an atomic number that is decreased by two.

Time spent: 0 sec

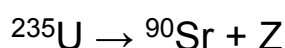
Subject:
General Chemistry

QID: 400199

System:
Atomic Theory and Chemical Composition

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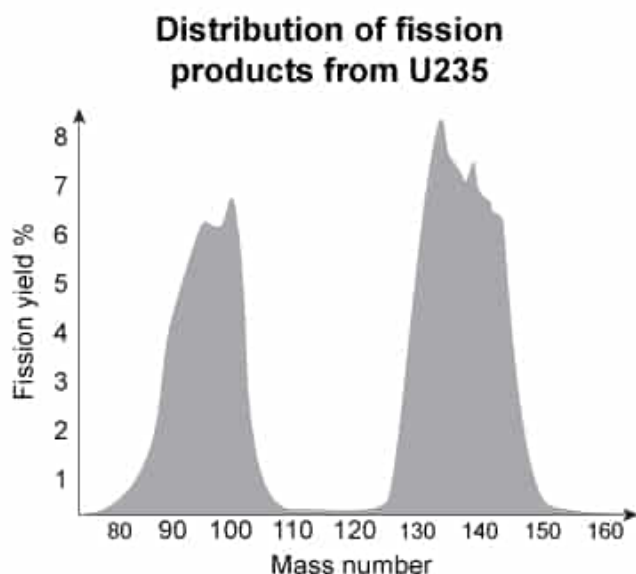


Figure 1 Distribution of ^{235}U fission products by mass number

To create the final ^{90}Y required for nuclear medicine studies, ^{90}Sr decay is carried out in a controlled ^{90}Y generator. ^{90}Y is then separated from residual ^{90}Sr for use in clinical applications.

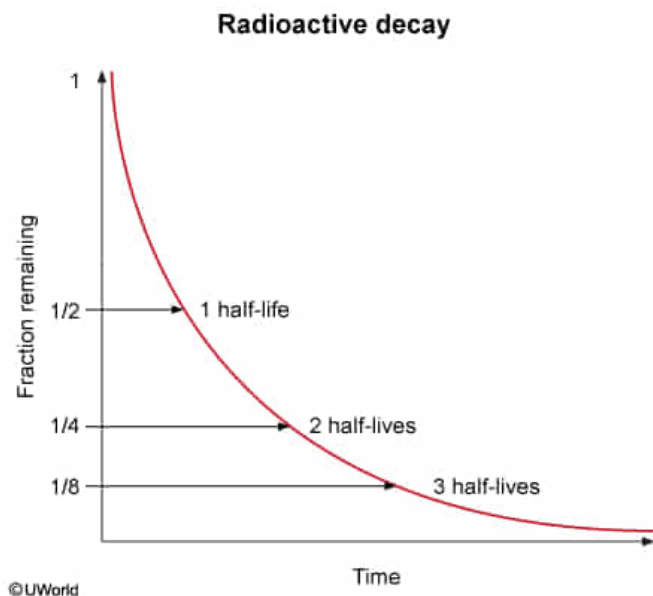
A patient is injected with 20 ng of ^{90}Y . Assuming the yttrium is not excreted, how long will it take for the initial amount of ^{90}Y to be reduced to 2.5 ng?

- ☐ A. 64 hours [1%]
- ☐ B. 128 hours [5%]
- ✓ ☒ C. 192 hours [85%]
- ☐ D. 256 hours [7%]

Correct

85% answered correctly

Explanation:



Radioactive decay is a type of exponential decay, meaning that a fixed proportion of the substance decays over any particular time interval. The rate is usually expressed in terms of radioactive **half-life** $t_{1/2}$, which is the amount of **time** needed for **half of the remaining substance to decay**. The fraction remaining after n half-lives can be calculated by:

$$\text{fraction remaining} = \frac{1}{2^n}$$

At one half-life (ie, $n = 1$), $\frac{1}{2}$ or 50% of the original substance remains; at two half-lives (ie, $n = 2$), $\frac{1}{4}$ or 25% remains; at three half-lives (ie, $n = 3$), $\frac{1}{8}$ or 12.5% remains; and so on.

The final quantity of 2.5 ng is $\frac{1}{8}$ of the initial quantity of ^{90}Y injected into the patient.

$$\frac{2.5 \text{ ng}}{20 \text{ ng}} = \frac{1}{8}$$

Per the passage, the half-life of ^{90}Y is 64 hours; therefore, its decay into $\frac{1}{8}$ of its initial quantity is:

$$\frac{1}{2^3} = \left(\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \right) = \frac{1}{8}$$

$$n = 3 \text{ half-lives}$$

$$3 \text{ half-lives} \times 64 \text{ hours} = 192 \text{ hours}$$

(Choices A, B, and D) The half-life of ^{90}Y is 64 hours. After 64 hours, 10 ng ($\frac{1}{2}$ or 50%) of the original substance remains; after 128 hours (two half-lives), 5 ng ($\frac{1}{4}$ or 25%) remains; and after 256 hours (four half-lives), 1.25 ng ($\frac{1}{16}$ or 6.25%) remains.

Educational objective:

For radioactive particles, the half-life $t_{1/2}$ is the amount of time needed for half the quantity of a substance to decay. The fraction remaining after a number of half-lives can be calculated using a formula for exponential decay, $1/2^n$, where n is the number of elapsed half-lives.

Time spent: 0 sec

Subject:
General Chemistry

QID: 400200

System:
Atomic Theory and Chemical Composition